DELUX MART: AN INTELLIGENT AI-POWERED E-COMMERCE PLATFORM FOR PERSONALIZED ONLINE SHOPPING

A PROJECT REPORT

Submitted in the partial fulfillment of requirements to

R. V. R & J. C. COLLEGE OF ENGINEERING

For the award of the degree
B. Tech. in CSE

By

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MAY, 2026

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CERTIFICATE

This is to certify that this project work titled **"DELUX MART: AN INTELLIGENT AI-POWERED E-COMMERCE PLATFORM FOR PERSONALIZED ONLINE SHOPPING"** is the work done by Oruganti Monik Paparao(Y22CS139), Pendyala Skanda Bhagavan(Y22CS145) and Tulam Sai Sudheer (Y22CS184) under our supervision, and submitted in partial fulfilment of the requirements for the award of the degree, B. Tech. in Computer Science and Engineering, during the Academic Year **2025-2026.**

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ACKNOWLEDGEMENT

The successful completion of any task would be incomplete without proper suggestions, guidance and support. Combination of these factors acts like backbone to our project titled **"DELUX MART: AN INTELLIGENT AI-POWERED E-COMMERCE PLATFORM FOR PERSONALIZED ONLINE SHOPPING".**

We are very glad to express our special thanks to **Smt. N. Zareena**, Project Guide, who has inspired us to select this topic, and also for her valuable suggestion in completion of the project.

We are very thankful to **Dr. S J R K Padminivalli V**, Project In-charge who extended her encouragement and assistance in ensuring the smooth progress of the project work.

We would like to express our profound gratitude to **Dr. M. Sreelatha**, Head of the Department of Computer Science and Engineering for her encouragement and support to carry out this Project successfully.

We are very much thankful to **Dr. Kolla Srinivas,** Principal, for having allowed us to conduct the study needed for the project.

Finally we submit our heartfelt thanks to all the staff in the Department and to all our friends for their cooperation during the project work.

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ABSTRACT

The rapid growth of e-commerce platforms has created a need for smart, scalable, and easy-to-use technologies that improve customer experience and decision-making. This study presents the idea and implementation of DeLux Mart, a web-based e-commerce platform that integrates advanced features and modern technologies to provide a seamless shopping experience. The system offers a complete online shopping environment, including efficient order processing, product browsing, cart management, secure user authentication, and seller-side functionalities.

A key highlight of this project is the inclusion of an AI-powered recommendation engine, which enhances personalization and user engagement. By analyzing product data, user behavior, and preferences, the system intelligently suggests relevant items, helping users discover products more easily and improving overall satisfaction. This personalized approach not only benefits customers but also increases sales opportunities for sellers.

In addition, the platform incorporates a generative AI-powered intelligent query system that allows users to interact with the application in a more natural and conversational manner. This feature enables users to search for products, ask questions, and receive meaningful responses quickly, thereby improving usability and accessibility.

The frontend of the system is designed using responsive web technologies to ensure compatibility across multiple devices, including desktops, tablets, and smartphones. On the backend, robust server-side technologies and API integrations are used to handle data processing, communication, and system operations efficiently. The platform also utilizes structured databases to manage product information, user data, and transaction records in an organized and secure manner. The system emphasizes reliability and performance through extensive testing modules, validation techniques, and error-handling mechanisms.

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S.No	Abbreviation	Full Form
1.	NLP	Natural Language Processing
2.	LR	Logistic Regression
3.	TF-IDF	Term Frequency-Inverse Document Frequency
4.	HCI	Human-Computer Interaction
5.	HLD	High-Level Design
6.	LLD	Low-Level Design
7.	LLM	Large Language Model
8.	MAP	Mean Average Precision
9.	MAR	Mean Average Recall
10.	NCF	Neural Collaborative Filtering
11.	CUDA	Compute Unified Device Architecture
12.	R@K	Recall at K
13.	ROC	Receiver Operating Characteristic
14.	UML	Unified Modeling Language